A 94-GHz 6-bit High-Precision Phase Shifter Based on Dual-Injection Gilbert Vector Modulator

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***Abstract*— This article presents a 94-GHz 6-bit high-precision phase shifter based on a dual-injection Gilbert vector modulator (DIG-VM). It is composed of two polarity switching amplifiers (PSAs), an *I*/*Q* signal generator, and a DIG-VM. The proposed DIG-VM not only maintains a constant input impedance in all phase states, thus suppressing the loading effect of the *I*/*Q* signal generator but also reduces the maximum errors caused by *I*/*Q* signal mismatches. Theoretical analysis shows that the maximum phase error due to *I*/*Q* signal phase mismatch and the maximum gain error due to *I*/*Q* signal magnitude mismatch are only half of those in the conventional Gilbert VM. In addition, the 90° PSA and 180° PSA are designed to complete the 360° phase shift and provide additional gain. The proposed phase shifter is fabricated using 0.13-μm SiGe BiCMOS process with a core area of 750 x 450 μm. The measured results show that the proposed phase shifter achieves an rms gain error of 0.58∼0.85 dB and an rms phase error of 1.2° ∼2.9° across 91.5∼98 GHz. In addition, the peak average gain reaches 14.4 dB at 94.5 GHz and the input 1-dB compression point is −14 dBm at 94 GHz.**

***Index Terms*— Constant impedance, dual-injection Gilbert vector modulator (DIG-VM), high precision, millimeter wave (mm-wave), phase shifter, SiGe BiCMOS, vector modulation.**

I. Introduction

RECENTLY, phased array technology has made significant progress due to advancements in millimeter-wave communication systems and autonomous radars. Millimeter-wave phased arrays achieve electronic beamforming and scanning by adjusting the phase of the electromagnetic (EM) waves transmitted by the antenna elements. This technology offers advantages such as improved signal-to-noise ratio (SNR) and effective isotropic radiated power (EIRP), and has found widespread applications in radar imaging [1], [2], [3], [4] and high-speed communication [5], [6], [7]. The phase shifter is one of the core modules of phased array chips, playing a crucial role in signal phase adjustment.

Phase shifters are primarily classified into passive and active structures. Passive phase shifters include switch type (STPS) [8], [9], [10] and reflective type (RTPS) [11], [12], [13], which offer advantages such as no static power consumption, high accuracy, and high linearity. However, they exhibit high noise figure, large insertion loss, and relatively large areas. Active phase shifters based on the principle of vector modulation, featuring high gain, compact size, and easy phase control, have garnered widespread attention and research interest [14]. These phase shifters are primarily composed of an *I*/*Q* signal generator and a vector modulator (VM). It is important that the precision of the output signal of VM is highly dependent on the accuracy of the *I*/*Q* signals. Therefore, improving the performance of the phase shifter requires high-quality *I*/*Q* signals.

Quadrature all-pass filters (QAFs) using the resistor and inductor compensation networks have been proposed to generate *I*/*Q* signals with low amplitude and phase mismatches [15] and [16]. Furthermore, a novel *RC* multiphase filter (PPF) has been presented to further improve the accuracy of *I*/*Q* signals [17]. However, all of these *I*/*Q* signal generators require a specific load impedance. Unfortunately, the conventional Gilbert VM is achieved by changing the gain of the *I*/*Q* paths amplifiers and their input impedance varies significantly across different phase states. This causes a challenge for the design of high-precision phase shifters.

To address this issue, a 94-GHz 6-bit high-precision phase shifter based on dual-injection Gilbert VM (DIG-VM) is proposed. Theoretical analysis of the DIG-VM shows that the proposed DIG-VM maintains a constant input impedance across all phase states, thus improving the quality of the *I*/*Q* signals. Moreover, it minimizes the maximum errors caused by *I*/*Q* signal mismatches compared to the conventional VM and achieves a high-precision phase shift at 94 GHz.

The structure of this article is organized as follows. Section II presents the architecture of the proposed high-precision phase shifter. Section III describes the principle of the DIG-VM along with analyses of the input impedance and the impact of *I*/*Q* signal mismatches. Section IV gives a detailed description of the circuit implementation and simulation results. The measurement results are depicted and discussed in Section V. Finally, the conclusions are presented in Section VI.

II. Architecture of the High-Precision Phase Shifter

The architecture of the proposed high-precision phase shifter is shown in Fig. 1, including a 90° polarity switching

Received 13 December 2024; revised 29 January 2025; accepted 20 February 2025. Date of publication 4 March 2025; date of current version 15 August 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62074121, Grant 62034002, Grant 62021004, and Grant 62274133. *(Corresponding author: Zhangming Zhu.)*

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Digital Object Identifier 10.1109/TMTT.2025.3545028

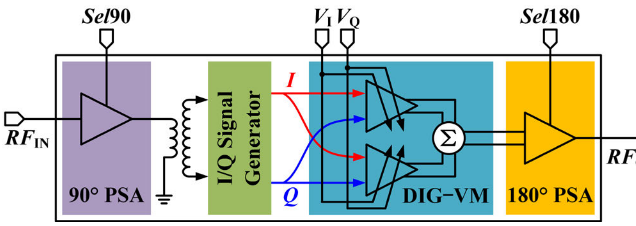

Fig. 1. Architecture of the proposed 94-GHz 6-bit high-precision phase shifter.

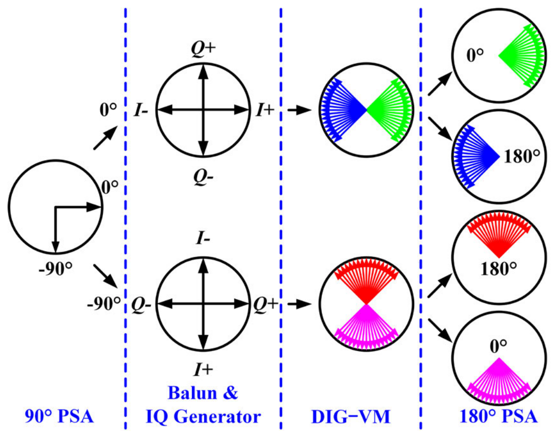

Fig. 2. Vector modulation principle of the proposed phase shifter.

amplifier (PSA) (90° PSA), an I/Q signal generator, a DIG-VM, and a 180° PSA. The DIG-VM provides a constant loading impedance for the I/Q signal generator, ensuring the accuracy of the I/Q signals in different phase states. With high-quality I/Q signals and minimal impact of I/Q signal mismatches, the proposed DIG-VM enables high-precision vector modulation within one quadrant of the phase plane.

To extend the high-precision phase shift to a 360° range, the 90° PSA and the 180° PSA are introduced to achieve phase shift of 90° and 180°, respectively. The 180° PSA, which consists of two symmetrical cascode amplifiers, is connected to the differential output of the DIG-VM, to select the phase state by adjusting the transistor voltages. It removes the need for a balun, thereby saving valuable chip area. Based on the 180° PSA, the 90° PSA inserts a phase shift network between the common emitter (CE) and common base (CB) transistors in the phase shift path amplifier to realize a 90° phase shift. The 90° PSA and 180° PSA provide both phase shift and gain, compensating for the insertion loss of the passive networks in the phase shifter.

As a result, the phase shifter achieves high-precision and high-gain phase shifts across the entire 360° range. The vector modulation principle is shown in Fig. 2.

III. Principle and Analysis of the DIG-VM

To avoid the degradation of I/Q signal accuracy caused by impedance variations, the DIG-VM with a constant input impedance across all phase states is proposed. Moreover, the dual-injection Gilbert vector modulator (DIG-VM) reduces the maximum errors of the output caused by I/Q signal mismatches, thereby improving the phase shift accuracy of the phase shifter.

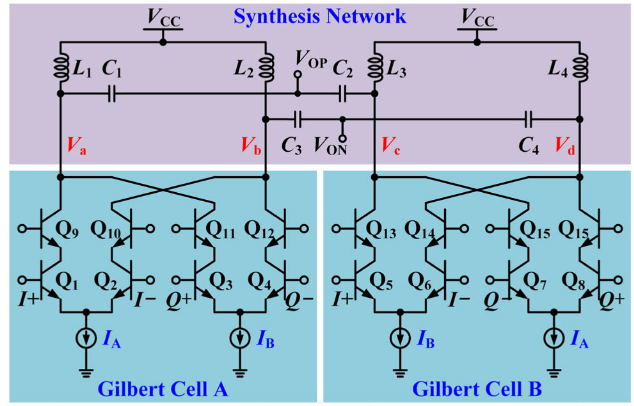

Fig. 3. Structure of the proposed DIG-VM.

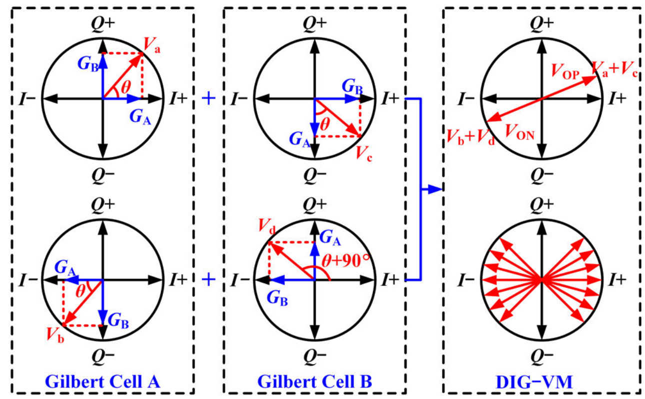

Fig. 4. Vector modulation principle of the proposed DIG-VM.

A. Principle of the DIG-VM

The structure of the proposed DIG-VM shown in Fig. 3 consists of two identical Gilbert cells, which have a dual relationship in signal synthesis and circuit injection. In Gilbert cell A, the I path injects a current of I_A, while the Q path injects a current of I_B. In contrast, the I path injects a current of I_B and the Q path injects a current of I_A in Gilbert cell B. The sum of I_A and I_B remains constant. The two Gilbert cells are composed of cascode amplifiers. The bases of all CE transistors are connected to a common dc voltage, while the bases of all CB transistors are connected to a different common dc voltage. Therefore, all differential paths equally split the tail current.

Fig. 4 indicates the vector modulation principle of the DIG-VM. $I+$, $I-$, $Q+$, and $Q-$ represent the quadrature input signals, while G_A and G_B denote the gains with currents $I_A/2$ and $I_B/2$, respectively. The Gilbert cell A and Gilbert cell B modulate the input I/Q signals by controlling the injection currents I_A and I_B, resulting in two pairs of differential signals, V_a and V_b, as well as V_c and V_d, which have equal amplitudes and a phase difference of 90°. The synthesis network combines the signals V_a and V_c as well as V_b and V_d to generate the final pair of differential output signals, V_{OP} and V_{ON}.

B. Input Impedance Analysis

According to Miller's theorem, the input admittance of a cascode amplifier can be represented as

$$Y_{\text{in}} = sC_\pi + s(1 - \mathrm{A}_v)C_\mu + \frac{1}{r_\pi} \tag{1}$$

where A_v is the gain of the CE transistor. For simplicity in analysis, A_v is approximated as -1. For HBTs, $C_\mu \ll C_\pi$, Y_{in} can be simplified as

$$Y_{\text{in}} = s\left(C_\pi + 2C_\mu\right) + \frac{1}{r_\pi} \approx sC_\pi + \frac{1}{r_\pi}. \tag{2}$$

C_π and $1/r_\pi$ are

$$C_\pi = \frac{\mathrm{d}Q_{\mathrm{F}}}{\mathrm{d}V_{\mathrm{BE}}} + C_{\text{je}} = \tau_{\mathrm{F}}\frac{I_{\mathrm{C}}}{V_{\mathrm{T}}} + C_{\text{je}} \tag{3}$$

$$\frac{1}{r_\pi} = \frac{\mathrm{d}I_{\mathrm{B}}}{\mathrm{d}V_{\mathrm{BE}}} = \frac{1}{\beta_{\mathrm{F}}}\frac{I_{\mathrm{C}}}{V_{\mathrm{T}}} \tag{4}$$

where Q_{F} is the charge stored in the base region, τ_{F} is the forward transit time, I_{C} is the collector current, V_{T} is the thermal voltage, β_{F} is the forward current gain, and C_{je} is the BE junction depletion layer capacitance, which is independent of I_{C}. Combining (2)–(4), it can be concluded that the input admittance of the cascode amplifier is linearly related to I_{C}.

The conventional Gilbert VM achieves vector modulation by changing the currents in the I/Q path amplifiers. Therefore, their input impedance varies significantly under different phase states. This will produce a serious loading effect for the I/Q signal generator across all phase states, degrading the quality of I/Q signals, thus impacting the accuracy of the phase shift obviously [18].

For the DIG-VM shown in Fig. 3, Gilbert cell A and Gilbert cell B exhibit a dual state to achieve vector modulation. The input admittance of the I/Q paths is the sum of the input admittances of the two Gilbert cells and can be expressed by

$$\begin{aligned} Y_{\text{in}} &= sC_{\pi\mathrm{A}} + \frac{1}{r_{\pi\mathrm{A}}} + sC_{\pi\mathrm{B}} + \frac{1}{r_{\pi\mathrm{B}}} \\ &= s\left(C_{\text{jeA}} + C_{\text{jeB}} + \tau_{\mathrm{F}}\frac{I_{\mathrm{A}} + I_{\mathrm{B}}}{V_{\mathrm{T}}}\right) + \frac{I_{\mathrm{A}} + I_{\mathrm{B}}}{\beta_{\mathrm{F}}V_{\mathrm{T}}} \end{aligned} \tag{5}$$

where $I_{\mathrm{A}} + I_{\mathrm{B}}$ is always constant. Therefore, the input admittance of the I/Q paths in the DIG-VM remains constant at all phase states.

Fig. 5(a) and (b) shows the input impedance variations of the real and imaginary parts of the DIG-VM I/Q paths at $I_{\mathrm{A}} + I_{\mathrm{B}} = 5$ mA, along with a comparison to the conventional Gilbert VM. At 94 GHz, the input impedance of the DIG-VM I/Q paths remains almost constant, with the real and imaginary parts changing only by 6 and 8 Ω. These variations are only 10% of those observed in the conventional Gilbert VM, effectively suppressing the loading effect of the I/Q signal generator and improving the phase shift accuracy.

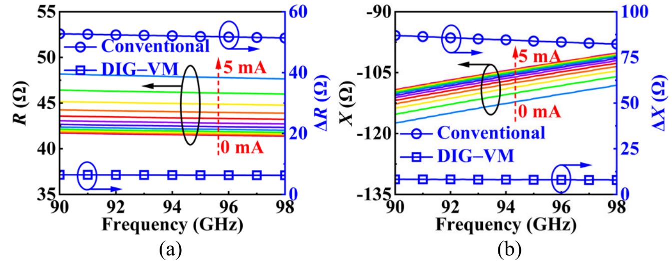

Fig. 5. Simulation results of (a) real and (b) imaginary parts of the input impedance of the DIG-VM and conventional Gilbert VM I/Q paths.

C. Gain and Phase Error Analysis

The I/Q signals with mismatches generated by the I/Q signal generator are given as

$$\begin{cases} I+ = \mathrm{A}e^{\mathrm{j}0} \\ Q+ = \mathrm{A}(1+\Delta\mathrm{A})e^{\mathrm{j}(\pi/2+\Delta\theta)} \\ I- = \mathrm{A}e^{\mathrm{j}\pi} \\ Q- = \mathrm{A}(1+\Delta\mathrm{A})e^{\mathrm{j}(3\pi/2+\Delta\theta)} \end{cases} \tag{6}$$

where A represents the amplitude of the I/Q signals, $\Delta\mathrm{A}$ is the amplitude mismatch error, and $\Delta\theta$ is the phase mismatch error. The inductors and capacitors in the I/Q path-matching network are identical, so their specific effects are neglected.

Assuming that there are no additional errors during 90° and 180° phase shifting, the errors caused by I/Q signal mismatches across the 360° range can be analyzed by considering the errors within one quadrant. An analysis of the errors of the V_{OP} synthesized by I/Q signals with mismatches is performed. V_{a} and V_{c} with I/Q signal mismatches are expressed as

$$\begin{cases} V_{\mathrm{a}} = G_{\mathrm{A}}(I+) + G_{\mathrm{B}}(Q+) \\ = AG_{\mathrm{A}} + \mathrm{A}(1+\Delta\mathrm{A})G_{\mathrm{B}}\left[-\sin(\Delta\theta) + \mathrm{j}\cos(\Delta\theta)\right] \\ V_{\mathrm{c}} = G_{\mathrm{B}}(I+) + G_{\mathrm{A}}(Q-) \\ = AG_{\mathrm{B}} - \mathrm{A}(1+\Delta\mathrm{A})G_{\mathrm{A}}\left[-\sin(\Delta\theta) - \mathrm{j}\cos(\Delta\theta)\right] \end{cases} \tag{7}$$

V_{a} and V_{c} are then synthesized in V_{OP} and can be represented by

$$\begin{aligned} V_{\mathrm{OPr}} &= V_{\mathrm{a}} + V_{\mathrm{c}} \\ &= \mathrm{A}[(G_{\mathrm{A}} + G_{\mathrm{B}}) - (1+\Delta\mathrm{A})(G_{\mathrm{B}} - G_{\mathrm{A}})\sin(\Delta\theta)] \\ &\quad + \mathrm{jA}(1+\Delta\mathrm{A})(G_{\mathrm{B}} - G_{\mathrm{A}})\cos(\Delta\theta). \end{aligned} \tag{8}$$

V_{OP} with ideal I/Q signals is

$$V_{\mathrm{OPi}} = \mathrm{A}(G_{\mathrm{A}} + G_{\mathrm{B}}) + \mathrm{jA}(G_{\mathrm{B}} - G_{\mathrm{A}}). \tag{9}$$

From (8) and (9), the phase error between the ideal signal and the real signal is

$$\begin{aligned} \theta_e &= |\theta_r - \theta_i| \\ &= \left| \begin{matrix} \arctan \frac{(1+\Delta\mathrm{A})(G_{\mathrm{B}}-G_{\mathrm{A}})\cos(\Delta\theta)}{(G_{\mathrm{A}}+G_{\mathrm{B}})-(1+\Delta\mathrm{A})(G_{\mathrm{B}}-G_{\mathrm{A}})\sin(\Delta\theta)} \\ -\arctan \frac{G_{\mathrm{B}}-G_{\mathrm{A}}}{G_{\mathrm{A}}+G_{\mathrm{B}}} \end{matrix} \right| \end{aligned} \tag{10}$$

where θ_r and θ_i are the phases of V_{OPr} and V_{OPi}, respectively.

When the I/Q signals only have a phase mismatch, $\Delta\mathrm{A} = 0$. Equation (10) can be rewritten as

$$\theta_e = \left| \arctan \frac{\cos(\Delta\theta)}{\frac{G_{\mathrm{A}}+G_{\mathrm{B}}}{G_{\mathrm{B}}-G_{\mathrm{A}}} - \sin(\Delta\theta)} - \arctan \frac{G_{\mathrm{B}} - G_{\mathrm{A}}}{G_{\mathrm{A}} + G_{\mathrm{B}}} \right|. \tag{11}$$

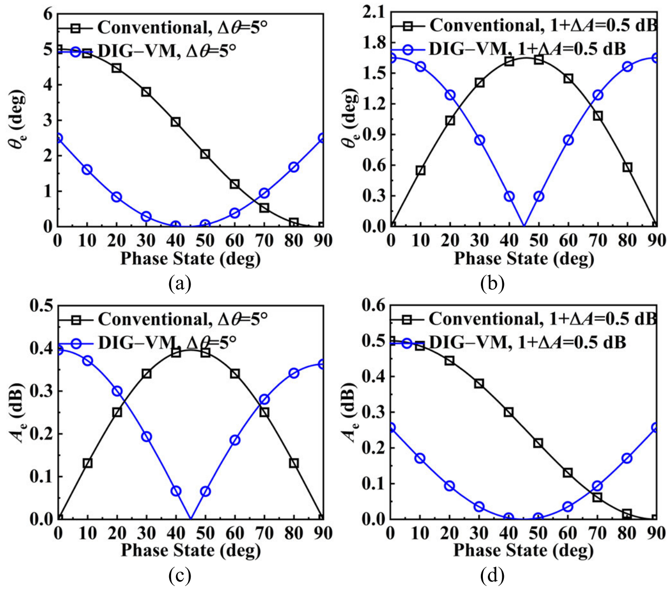

Fig. 6. Phase errors due to I/Q signal (a) phase mismatch and (b) magnitude mismatch, and gain errors due to I/Q signal (c) phase mismatch and (d) magnitude mismatch across different phase states.

It can be seen that the maximum phase error is $\Delta\theta/2$, while the conventional Gilbert VM has a maximum error of $\Delta\theta$ [19]. The maximum phase error of the DIG-VM is only half of that of the conventional Gilbert VM. The detailed derivation process is shown in the Appendix. Fig. 6(a) shows the maximum phase errors under different phase states when $\Delta\theta = 5°$ and $\Delta A = 0$. Fig. 6(a) shows the phase errors under different phase states when $\Delta\theta = 5°$ and $\Delta A = 0$. The phase error is zero at the phase state of 45° and reaches its maximum value of $\Delta\theta/2$ at the phase states of 0° and 90°. Meanwhile, the phase errors of the DIG-VM are smaller than those of the conventional Gilbert VM for phase states ranging from 0° to 67°.

When the I/Q signals only have a magnitude mismatch, $\Delta\theta = 0$. Equation (10) can be denoted as

$$\theta_e = \left|\arctan\left[(1+\Delta A)\frac{G_A - G_B}{G_A + G_B}\right] - \arctan\frac{G_A - G_B}{G_A + G_B}\right|. \quad (12)$$

In general, the amplitude mismatch error of the I/Q signals satisfies $\Delta A \ll 1$. According to Taylor's theorem, θ_e can be reduced to

$$\theta_e \approx \left|\Delta A \frac{G_A - G_B}{G_A + G_B} \bigg/ \left[1 + \left(\frac{G_A - G_B}{G_A + G_B}\right)^2\right]\right|. \quad (13)$$

It can be observed that the maximum value of θ_e in this case is $|\Delta A/2|$, which is the same as that of the conventional Gilbert VM. Fig. 6(b) shows the phase errors under different phase states when $\Delta\theta = 0$ and $1 + \Delta A = 0.5$ dB. Contrary to the conventional Gilbert VM, the phase error of the DIG-VM is zero at the phase state of 45° and reaches its maximum value at phase states of 0° and 90°.

Similarly, from (8) and (9), the gain error between the ideal signal and the real signal is

$$A_e = |20\log A_r/A_i|$$

$$= \left|10\log\frac{[G_A + G_B - (1+\Delta A)(G_B - G_A)\sin(\Delta\theta)]^2 + (1+\Delta A)^2(G_B - G_A)^2\cos^2(\Delta\theta)}{2(G_A^2 + G_B^2)}\right| \quad (14)$$

where A_r and A_i are the magnitudes of V_{OPr} and V_{OPi}, respectively.

When the I/Q signals only have a phase mismatch, $\Delta A = 0$. Equation (14) can be expressed as

$$A_e = \left|10\log\frac{[(G_A + G_B) - (G_B - G_A)\sin(\Delta\theta)]^2 + (G_B - G_A)^2\cos^2(\Delta\theta)}{2(G_A^2 + G_B^2)}\right|$$
$$= \left|10\log\left[1 - \frac{G_B^2 - G_A^2}{G_A^2 + G_B^2}\sin(\Delta\theta)\right]\right|. \quad (15)$$

The maximum gain error is |10log(1-sin|$\Delta\theta$|)|. The maximum gain error is the same as that of the conventional Gilbert VM. Fig. 6(c) shows the gain errors under different phase states when $\Delta\theta = 5°$ and $\Delta A = 0$. Contrary to the conventional Gilbert VM, the gain error of the DIG-VM is zero at the phase state of 45° and reaches its maximum value at phase states of 0° and 90°, similar to the case in Fig. 6(b).

When the I/Q signals only have a magnitude mismatch, $\Delta\theta = 0$. Equation (14) can be rewritten as

$$A_e = \left|10\log\frac{(G_A + G_B)^2 + (1+\Delta A)^2(G_B - G_A)^2}{2(G_A^2 + G_B^2)}\right|$$
$$= \left|10\log\frac{2(G_A^2 + G_B^2) + (2\Delta A + \Delta A^2)(G_B - G_A)^2}{2(G_A^2 + G_B^2)}\right|. \quad (16)$$

When $\Delta A \ll 1$, A_e can be simplified as

$$A_e \approx \left|10\log\left\{1 + \frac{(G_B - G_A)^2}{G_A^2 + G_B^2}\Delta A\right\}\right|. \quad (17)$$

The maximum gain error is |10log(1 + ΔA)|, whereas the maximum gain error of the conventional Gilbert VM is |10log(1 + 2ΔA)| ≈2|10log(1 + ΔA)|. In comparison, the maximum gain error of the DIG-VM is half of that of the conventional Gilbert VM. Fig. 6(d) shows the gain errors at different phase states when $\Delta\theta = 0$ and $1 + \Delta A = 0.5$ dB. Similar to the results of the phase errors caused by the I/Q signal phase mismatch, the gain error is zero at the phase state of 45° and reaches its maximum value of |10log(1 + ΔA)| at the phase states of 0° and 90°. Meanwhile, the DIG-VM exhibits a better performance for the phase states ranging from 0° to 67°.

To further compare the maximum phase errors and gain errors between the proposed DIG-VM and conventional Gilbert VM, the maximum errors due to both I/Q phase and magnitude mismatches are calculated. Fig. 7(a) and (b) compares the maximum phase errors while Fig. 7(c) and (d) compares the maximum gain errors. It can be observed that the proposed DIG-VM displays significant superiority in the

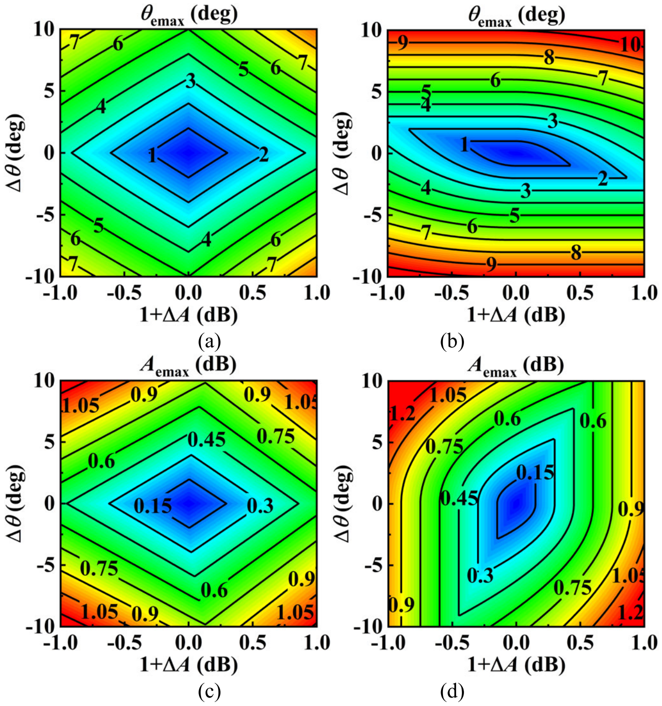

Fig. 7. Maximum phase errors of the (a) DIG-VM and (b) conventional Gilbert VM, and maximum gain errors of the (c) DIG-VM and (d) conventional Gilbert VM.

maximum error caused by both I/Q phase and magnitude mismatches, thereby the precision of the phase shifter can be improved greatly.

IV. Circuit Implement

The proposed high-precision phase shifter is fabricated in 0.13-μm SiGe BiCMOS technology, which provides seven metal layers, including two thick top metal layers (TM1 and TM2 layers). Moreover, the maximum transit/oscillation frequencies (i.e., f_T/f_{MAX}) for this process are 350/450 GHz. In order to reduce the transmission losses, all the inductors and transmission lines are implemented using TM2 or TM1 layer.

A. Balun and I/Q Signal Generator

The balun converts the single-ended signal into a differential signal for input to the I/Q signal generator and its layout is shown in Fig. 8(a). The balun adds a decoupling capacitor at the center tap. Furthermore, by adjusting the position of the center tap, the balun achieves a good balance. Within the 90∼98-GHz range, the EM simulated amplitude imbalance, phase imbalance, and insertion loss are less than 0.1 dB, 1°, and 2 dB, respectively, as shown in Fig. 8(b).

I/Q signal generators are usually accomplished by Lange coupler, coupled line, PPF, or QAF. Lange coupler and coupled line are composed of λ/4 transmission lines, offering low insertion loss but requiring larger areas. PPF consists of lumped components, occupying smaller areas but with higher insertion loss. Considering both insertion loss and layout area, QAF is selected for the I/Q signal generator, and its schematic and layout are shown in Fig. 9(a) and (b), respectively, where L_1

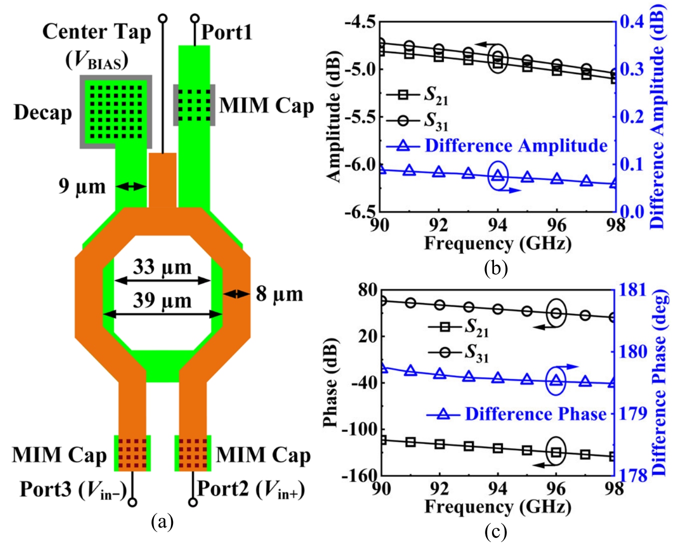

Fig. 8. (a) Layout and the EM simulation results of (b) amplitude and (c) phase imbalances of the balun.

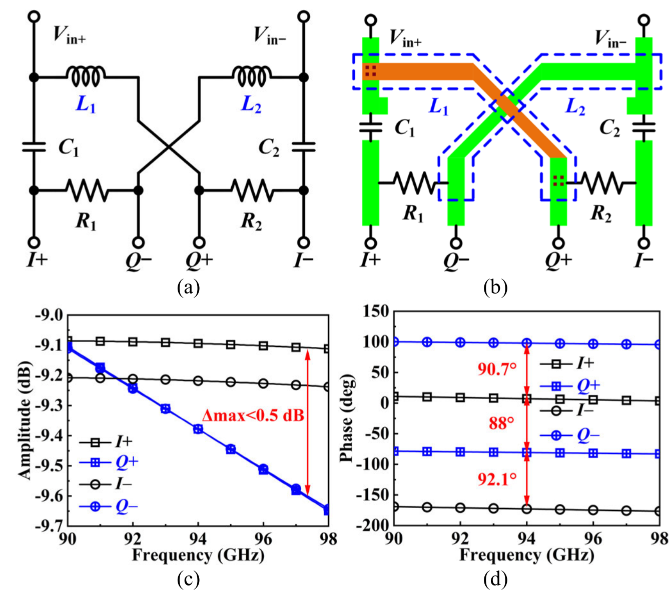

Fig. 9. (a) Schematic and (b) layout as well as the EM simulation results of (c) amplitude and (d) phase imbalance of the QAF.

and L_2 are 50 pH, C_1 and C_2 are 45 fF, and R_1 and R_2 are 80 Ω. As shown in Fig. 9(c) and (d), the EM simulated amplitude imbalance, phase imbalance, and insertion loss are less than 0.5 dB, 2.1°, and 3.6 dB, respectively, across the frequency range of 90∼98 GHz.

B. DIG-VM

The schematic of the proposed DIG-VM with the synthesis network layout is shown in Fig. 10. $Q_1 \sim Q_8$ constitute Gilbert cell A while $Q_9 \sim Q_{16}$ constitute Gilbert cell B.

$M_1 \sim M_4$ and $M_9 \sim M_{10}$ form the current mirror to provide current I_A for path A as well as $M_5 \sim M_8$ and $M_{11} \sim M_{12}$ form the current mirror to provide current I_B for path B. The current mirrors adopt a cascode structure to reduce channel modulation effects and improve the current replication

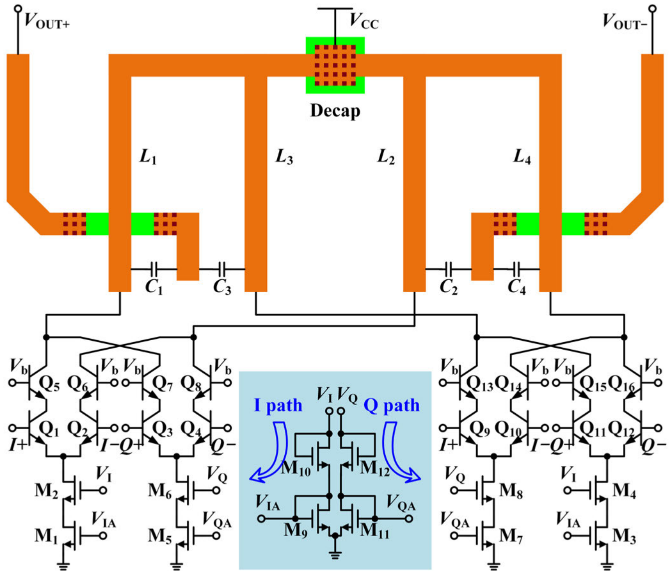

Fig. 10. Schematic of the DIG-VM with the synthesis network layout.

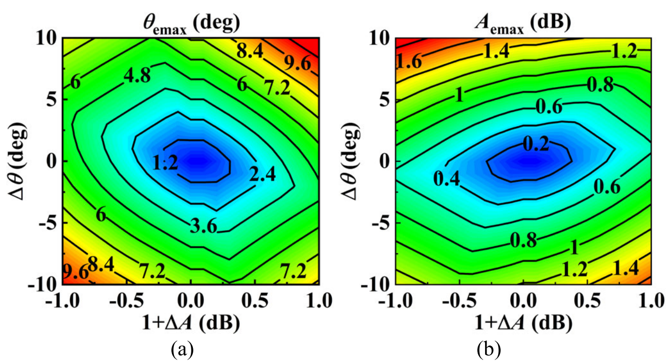

Fig. 11. Post-layout simulation results of the maximum (a) phase error and (b) gain error under I/Q signal mismatches of the proposed DIG-VM.

accuracy. The adjustments of I_A and I_B are achieved by tuning V_I and V_Q. $L_1 \sim L_4$ and $C_1 \sim C_2$ provide the output impedance matching and the synthesis of the signals. The sizes of $Q_1{\sim}Q_{16}$ and $M_1{\sim}M_8$ are 0.07 × 1.5 μm and 8/0.13 μm (W/L), respectively. The supply voltage V_{CC} is 3 V.

Although the proposed DIG-VM consumes more power compared to the conventional Gilbert VM, the signals generated by the two Gilbert cells are combined, resulting in an enhanced gain. The maximum phase and gain errors under I/Q signal mismatches of the DIG-VM are shown in Fig. 11(a) and (b), respectively. It is shown that the maximum phase and gain errors are slightly higher than the calculated results as shown in Fig. 7(a) and (c), respectively, due to the effects of high-frequency parasitics, but they still demonstrate excellent performance.

C. 180° PSA

The proposed 180° PSA consists of two symmetric cascode amplifiers, whose inputs are connected to the differential outputs of the DIG-VM, as shown in Fig. 12. The reference state or the phase shift state is selected by controlling the voltages of the two amplifiers. When V_{B0} and V_{C0} are equal to

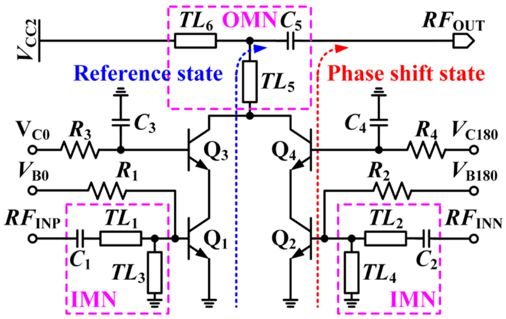

Fig. 12. Schematic of the 180° PSA.

V_{ONB} and V_{ONC} (bias voltages of $V_{ONB} = 0.95$ V and $V_{ONC} = 1.95$ V) while V_{B180} and V_{C180} are equal to V_{OFF} (bias voltage of $V_{OFF} = 0$ V), the reference state path works. Conversely, the phase shift path works. Since only one path is active, the two input impedances of the 180° PSA are unequal, which provides unbalanced differential loads for the DIG-VM. However, the use of a cascode structure in the DIG-VM improves the isolation between the output and input and mitigates the impact on the input impedance. The 180° PSA not only achieves phase shift but also provides additional gain. The 180° PSA has the potential to provide an excellent performance on phase and gain balance due to the fully symmetric structure. Furthermore, it eliminates the need for an on-chip balun for the differential to single conversion, saving valuable layout area. The size of $Q_1{\sim}Q_4$ is 0.07 × 1.5 μm, and the supply voltage V_{CC2} is 2 V. $TL_1 \sim TL_4$ as well as C_1 and C_2 constitute the input matching network (IMN), which ensures conjugate matching between the impedance of the active path and the output impedance of the DIG-VM. TL_5, TL_6, and C_5 constitute the output matching network (OMN), providing matching to a 50-Ω load.

The 180° PSA is simulated using a three-port configuration, with two input ports receiving signals of equal amplitude but opposite phases. Fig. 13(a) and (b) shows the post-layout simulated gain and phase imbalances of the 180° PSA, respectively, while Fig. 13(c) and (d) shows post-layout simulated input and output return losses, respectively. It can be observed that the gain reaches 8.9 dB at 95 GHz and the gain and phase imbalances are less than 0.1 dB and 1°, respectively, over the frequency range of 90∼98 GHz. The input and output return losses are both better than 10 dB across 90∼98 GHz.

D. 90° PSA

Based on the 180° PSA structure, the 90° PSA is realized by adding a 90° phase shift network in the phase shift path, as shown in Fig. 14. The phase shift network consists of TL_2, C_6, and C_7, where C_6 and C_7 are the parasitic capacitances of the collector of Q_2 and the emitter of Q_3, respectively. By adjusting the length of TL_2, a 90° phase shift can be achieved at 94 GHz. The size of $Q_1{\sim}Q_4$ is 0.07 × 1.5 μm, and the supply voltage V_{CC1} is 2 V.

Fig. 15(a) and (b) shows the post-layout simulated gain and phase imbalances between the reference and phase shift state, respectively. It can be observed that the gain and phase imbalances are less than 1 dB and 2.3°, respectively, within

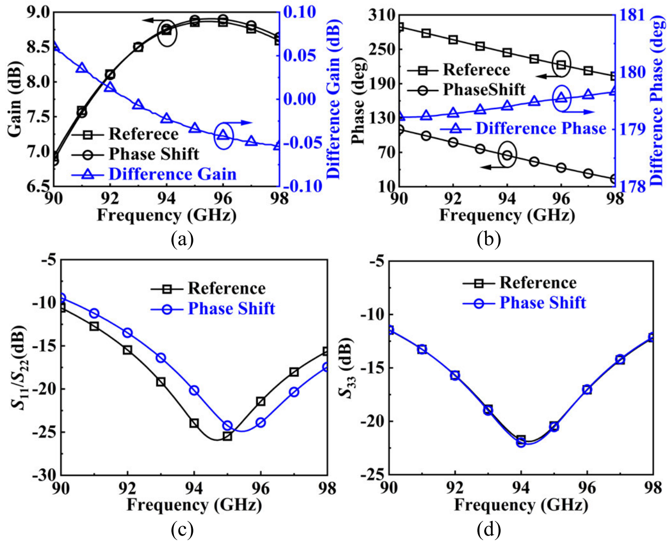

Fig. 13. Post-layout simulation results of (a) gain and (b) phase imbalances, as well as (c) S_{11}/S_{22} and (d) S_{33} of the 180° PSA.

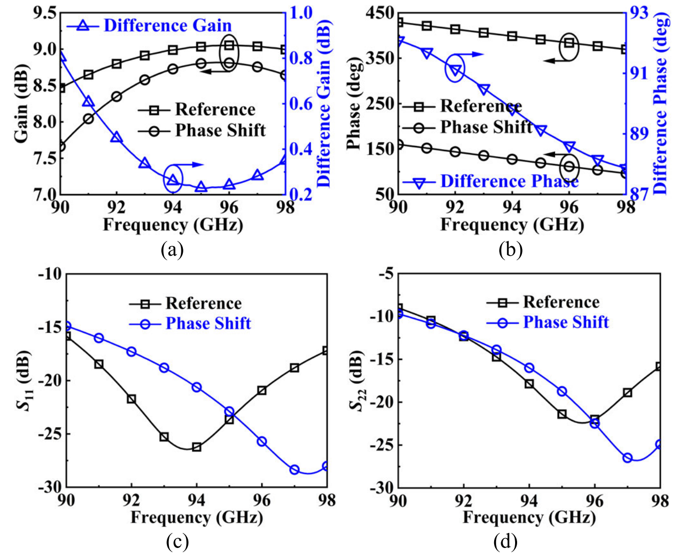

Fig. 15. Post-layout simulation results of (a) gain and (b) phase imbalance, and (c) S_{11} and (d) S_{22} of the 90° PSA.

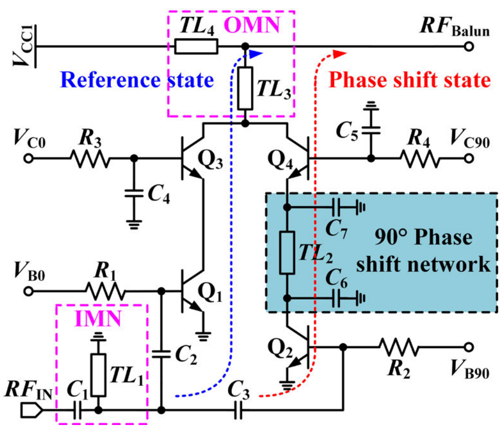

Fig. 14. Schematic of the 90° PSA.

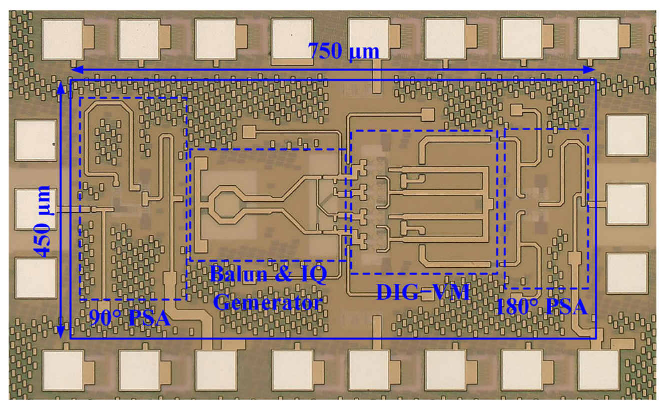

Fig. 16. Chip micrograph of the proposed phase shifter.

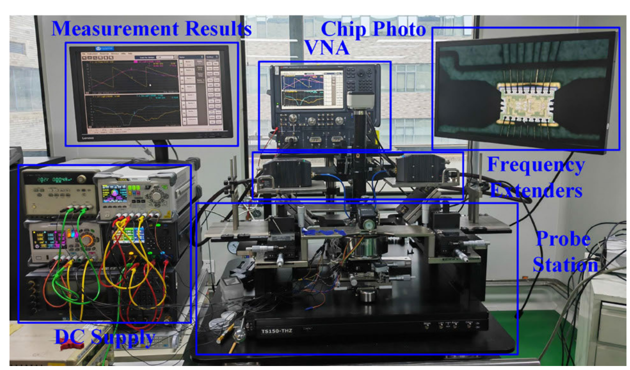

Fig. 17. Measurement setup for S parameters.

90∼98 GHz. Moreover, the gain reaches a peak of 8.9 dB at 95 GHz.

Unlike 180° PSA, the two paths of the 90° PSA are asymmetrical. The input and output impedance of the reference state and phase shift state are different. Therefore, TL_1, TL_3, and TL_4 are used as the matching network components to achieve good matchings for an average impedance of two states. As shown in Fig. 15(c) and (d), the post-layout simulated input and output return losses of both reference state and phase shift state are better than 15 and 10 dB, respectively.

V. Measurement Results

The micrograph of the fabricated chip is shown in Fig. 16 with an area of 750 × 450 μm^2. The proposed phase shifter consumes a dc power of 42 mW. The measurement setup is shown in Fig. 17. After standard short-open-load-through (SOLT) calibration, S parameters are measured on-chip using a vector network analyzer (VNA). The dc and ground pads are bonded to a printed circuit board (PCB) via chip-on-board (COB) bonding and the RF signal is delivered through 100-μm pitch ground-signal-ground (GSG) probes after passing through frequency extenders.

In the measurement, V_I and V_Q in the DIG-VM are adjusted by directly controlling the dc supply. The corresponding 64 voltage sets are obtained based on simulation results and calibrated according to the measurement results at 94 GHz. Furthermore, the bias voltages for the PSAs are switched using off-chip logic components.

TABLE I

SUMMARY OF PERFORMANCE AND COMPARISON WITH THE STATE-OF-THE-ART PHASE SHIFTERS

References	IMS 2024 [20]	MWTL 2024 [21]	TCASII 2024 [22]	TCASI 2023 [23]	TCASI 2021 [24]	TMTT 2020 [25]	TCASI 2018 [26]	JSSC 2017 [27]	**This Work**
Technology	65 nm bulk CMOS	28 nm CMOS	40 nm CMOS	65 nm CMOS	40 nm CMOS	65 nm CMOS	0.13 μm SiGe	28 nm DFSOI CMOS	**0.13 μm SiGe**
Frequency (GHz)	91~99	86~96	53~62	114~147	70~90	51~66.3	92~100	78.8~92.8	**91.5~98**
Resolution (bit)	6	6	6	6	6	5	5	4	**6**
Peak Average Gain (dB)	14.5***	-6.0	-7.0	-16.5	-15.1	-3.8	9.5	2.3	**14.4**
RMS Gain Error (dB)	0.2~1.0	0.18~0.47	0.4~0.55	0.6~1.5	0.6~1.2	0.25~0.72	1.4~1.8	1.5~2.0	**0.58~0.85**
RMS Phase Error (deg)	1.5~5.6	1.0~3.5	1.3~2.4	1.1~3.7	1.0~2.4	3.0~7.0	2.5~9.0	9.4~11.9	**1.2~2.9**
IP_{1dB} (dBm)	N/A	-12	N/A	N/A	N/A	-0.23	-17	-8~-5	**-14**
Noise Figure (dB)	N/A	N/A	N/A	N/A	N/A	17*	9.3~11**	9~14	**9.68***
P_{DC} (mW)	N/A	54	30	0	0	5	31	31.6	**42**
Core Chip Area (mm^2)	0.05	0.26	0.13	0.03	0.15	0.30	0.85	0.12	**0.33**
FOM1(GHz^2bits/°)	3835	751	542	879	626	390	1863	380	**6063**
FOM2(GHz^2bits/°)	N/A	N/A	N/A	N/A	N/A	1.5	0.1	0.31	**0.69**

*simulation result, **with LNA, ***with PA

$$\mathrm{FOM1}=\frac{f_0(\mathrm{GHz})\times G(\mathrm{lin.})\times B_{3dB}(\mathrm{GHz})\times \mathrm{Resolution}(\mathrm{bits})}{\theta_{\Delta,\mathrm{RMS},3dB}(\mathrm{deg})\times A_{\Delta,\mathrm{RMS},3dB}(\mathrm{lin.})},\quad \mathrm{FOM2}=\frac{f_0(\mathrm{GHz})\times G(\mathrm{lin.})\times B_{3dB}(\mathrm{GHz})\times IP_{1dB}(\mathrm{mW})\times \mathrm{Resolution}(\mathrm{bits})}{\theta_{\Delta,\mathrm{RMS},3dB}(\mathrm{deg})\times A_{\Delta,\mathrm{RMS},3dB}(\mathrm{lin.})\times P_{dc}(\mathrm{mW})\times (NF(\mathrm{lin.})-1)}$$

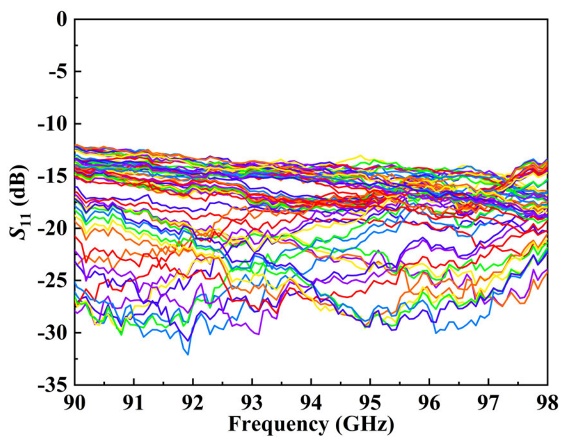

Fig. 18. The measurement results of S_{11} for 64 phase states.

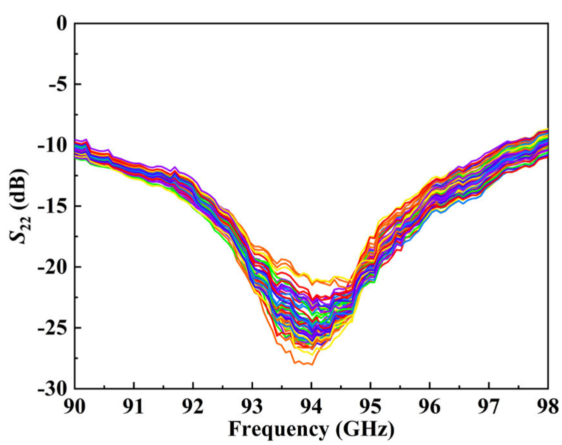

Fig. 19. Measurement results of S_{22} for 64 phase states.

The measured input and output return losses for the 64 states are shown in Figs. 18 and 19, respectively. It can be seen that in the 91.5∼98 GHz, the input and the output return losses are better than 13 and 10 dB, respectively.

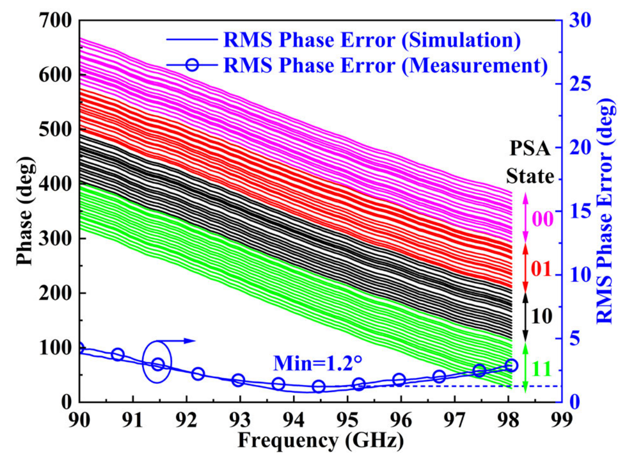

Fig. 20. Measurement and simulation results of the phase shift for 64 phase states and the rms phase error.

Fig. 20 shows the measured phase shift with 64 phase states and the calculated rms phase error. The phase shifter covers 360° phase shift range with a step size of 5.625° and the rms phase error is less than 2.9° within 91.5∼98 GHz, with a minimum value of 1.2° at 94.5 GHz. Fig. 21 presents the measured gain with 64 states and the calculated rms gain error. The average gain reaches a peak of 14.4 dB at 94.5 GHz, and the rms gain error is less than 0.85 dB across 91.5∼98 GHz with a minimum value of 0.58 dB at 94.5 GHz. The rms phase and gain errors are slightly higher than the post-layout simulation results. Potential causes for this discrepancy include transistor modeling inaccuracy, EM simulation inaccuracy, and process corner variations.

Fig. 22 shows the measured output power and the power gain of the phase shifter at 94 GHz with different input powers. It can be observed that the input P_{1dB} is −14 dBm. Due to the introduction of the 90° and 180° PSAs, the linearity of the phase shifter has deteriorated to conventional phase shifters.

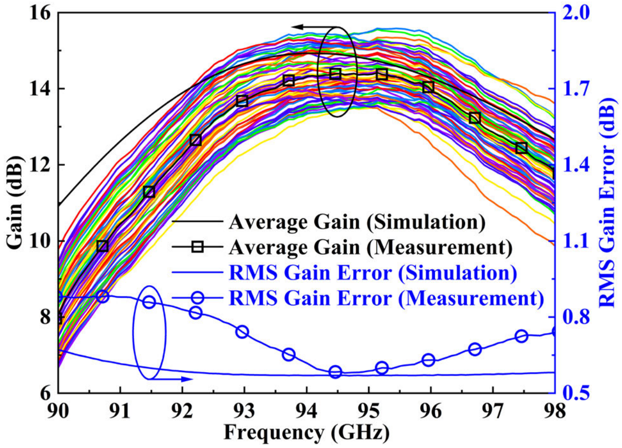

Fig. 21. Measurement and simulation results of the gain for 64 phase states and the average gain as well as the rms gain error.

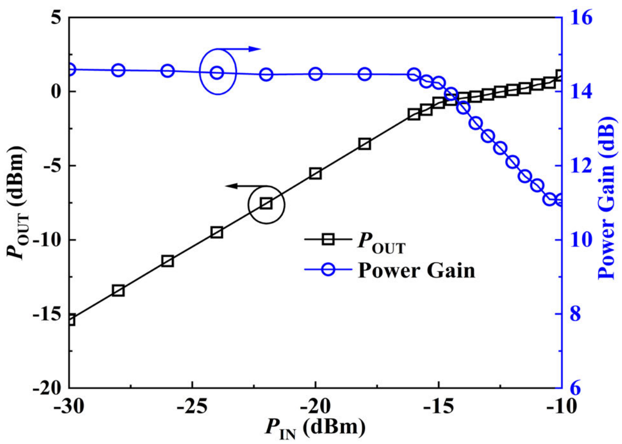

Fig. 22. Measurement results of the output power and power gain versus the input power.

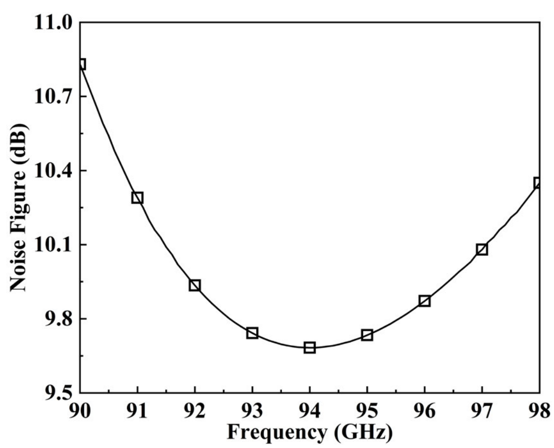

Fig. 23. Simulation results of the noise figure versus the frequency.

Fig. 23 shows the simulated noise figure of the phase shifter versus frequency. The noise figure is 9.68 dB at 94 GHz.

Table I shows the performance of the proposed phase shifter and a comparison with various phase shifters. It can be seen that the proposed 6-bit high-precision phase shifter achieves an excellent phase and gain performance with a competitive noise figure. Also, it exhibits high figures of merit (FOM1 and FOM2) among those works.

VI. Conclusion

In this article, a 6-bit high-precision phase shifter operating at 94 GHz is implemented using a 0.13-μm SiGe BiCMOS process. To reduce the loading effect of the I/Q signal generator and minimize the maximum errors caused by I/Q signal mismatches, a DIG-VM is proposed to achieve vector modulation. The DIG-VM keeps a constant input impedance of the I/Q paths in all phase states. Moreover, it experiences a half-maximum phase error due to I/Q signal phase mismatch and a half-maximum gain error due to I/Q signal magnitude mismatch compared to the conventional Gilbert VM. To achieve phase shift over the 360° range and compensate for the passive losses, the 90° PSA and 180° PSA are included. Measured results show that the proposed phase shifter provides a full 360° phase shifting range with a 5.625° step and the rms phase error of 1.2°∼2.9° within 91.5∼98 GHz. Furthermore, the rms gain error reaches 0.58∼0.85 dB over the entire frequency range with a peak average gain of 14.4 dB at 94.5 GHz. In addition, the input 1-dB compression point is −14 dBm at 94 GHz. Overall, the phase shifter demonstrates excellent phase control and amplitude balance. It meets the application requirements for W-band phased array chipsets, helping to improve system gain and resolution.

Appendix
Derivations of Modulate Errors

When the I/Q signals only have phase errors, the phase error of the DIG-VM is expressed as

$$\theta_e = \left| \arctan \frac{\cos(\Delta\theta)}{\frac{G_A+G_B}{G_B-G_A} - \sin(\Delta\theta)} - \arctan \frac{G_B - G_A}{G_A + G_B} \right|. \quad \text{(A.1)}$$

For convenience analysis, let the term inside the absolute value to denoted as ψ and $t = (G_B - G_A)/(G_A + G_B)$, where $t \in (-\infty, -1] \cup [1, +\infty)$. The derivative of ψ with respect to t is

$$\frac{d\psi}{dt} = \left| \frac{[1-\cos(\Delta\theta)]t^2 - 2t\sin(\Delta\theta) + [1-\cos(\Delta\theta)]}{(t^2+1)[t^2 - 2t\sin(\Delta\theta) + 1]} \right|. \quad \text{(A.2)}$$

Let (A.2) equals 0, the solution is

$$t_{1,2} = \frac{\sin(\Delta\theta)}{1-\cos(\Delta\theta)} \pm \frac{\sqrt{2\cos(\Delta\theta)[1-\cos(\Delta\theta)]}}{1-\cos(\Delta\theta)} = \begin{cases} \dfrac{\sqrt{1+\cos(\Delta\theta)} \pm \sqrt{2\cos(\Delta\theta)}}{\sqrt{1-\cos(\Delta\theta)}}, \Delta\theta > 0 \\ \dfrac{-\sqrt{1+\cos(\Delta\theta)} \pm \sqrt{2\cos(\Delta\theta)}}{\sqrt{1-\cos(\Delta\theta)}}, \Delta\theta < 0. \end{cases} \quad \text{(A.3)}$$

When $\Delta\theta > 0$, as $\Delta\theta \to 0$, $t_1 \to 0$, and $t_2 \to +\infty$, ψ is monotonically increasing in the range of $(-\infty, -1]$ and monotonically decreasing in the range of $[1, +\infty)$. Similarly, when $\Delta\theta < 0$, ψ is monotonically decreasing in the range of $(-\infty, -1]$ and monotonically increasing in the range of $[1, +\infty)$. From (A.1), it can be seen that $\theta_e \to 0$ as $t \to \infty$. Therefore, θ_e has a local maximum value at $t = -1$ or $t = 1$.

When $t = 1$, θ_e can be expressed as

$$\begin{aligned}\theta_e &= \left|\arctan \frac{\cos(\Delta\theta)}{1-\sin(\Delta\theta)} - \arctan 1\right| \\ &= \left|\arctan \frac{2\cos\left(\frac{\Delta\theta}{2}+\frac{\pi}{4}\right)\sin\left(\frac{\Delta\theta}{2}+\frac{\pi}{4}\right)}{2\cos^2\left(\frac{\Delta\theta}{2}+\frac{\pi}{4}\right)} - \frac{\pi}{4}\right| \\ &= \frac{\Delta\theta}{2}. \end{aligned} \tag{A.4}$$

When $t = -1$, θ_e can be expressed as

$$\begin{aligned}\theta_e &= \left|\arctan \frac{\cos(\Delta\theta)}{-1-\sin(\Delta\theta)} - \arctan(-1)\right| \\ &= \left|\arctan \frac{2\sin\left(\frac{\Delta\theta}{2}+\frac{\pi}{4}\right)\cos\left(\frac{\Delta\theta}{2}+\frac{\pi}{4}\right)}{-2\sin^2\left(\frac{\Delta\theta}{2}+\frac{\pi}{4}\right)} + \frac{\pi}{4}\right| \\ &= \frac{\Delta\theta}{2}. \end{aligned} \tag{A.5}$$

In summary, when the I/Q signals only have phase mismatch errors, the maximum phase error is $\Delta\theta/2$.

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